

EMMANUEL HARTMAN

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CURRENT POSITION

University of Houston

Houston, Texas

Aug. 2024 - Present

Post Doctoral Scholar

EDUCATION

Florida State University

Tallahassee, Florida

Aug. 2018 - May 2024

PhD - Mathematics

GPA: 3.853

University of North Carolina at Asheville

Asheville, North Carolina

Aug. 2014 - May. 2018

Bachelor of Arts in Pure Mathematics

with a Minor in Computer Science

and Distinction as a University Scholar

GPA: 3.938 (Summa Cum Laude); Mathematics GPA: 4.0; Computer Science GPA: 4.0

EXTENDED RESEARCH STAYS

Johns Hopkins University

Baltimore, Maryland

Oct. 2021 - Nov. 2021

Visiting Scholar at the Department of Applied Mathematics and Statistics

and Center for Imaging Sciences

Institut Henri Poincaré

Paris, France

Oct. 2022 - Nov. 2022

Participant of the Thematic Semester on Geometry and Statistics in Data Sciences

PUBLICATIONS

In Preparation

[20] M. Yin*, E. Hartman*, Z. Ahmad*, Y. Sukurdeep, S. Chen, J. Noh, L. Tran, N. Charon, M. Maggioni, E. Kholmovski, R. Bergers, D. Spragg, H. Calkins, N. Trayanova. Shape analysis reveals cardiac morphological difference in cardiovascular patients.

[19] Samundra Regmi, Emmanuel Hartman, Andreas Mang, and Nicolas Charon. Numerical Integration of Geodesic Shooting Equations for Invariant Sobolev Metrics on Curve and Surface Spaces

[18] Murad Hossen, Emmanuel Hartman, and Nicolas Charon. Nonlinear Classification and Regression in Geometric Measure Spaces

Preprints

[17] Chaima Zouaoui, Jonathan Cerqueira, Sylvain Arguillere, Martin Bauer, Nicolas Charon, Mohamed Daoudi, Emmanuel Hartman. Mesh-Agnostic Refinement Models for Latent Space Encoding of 3D Human Body Shapes.

Preprint available on request.

[16] *Gary P. T. Choi, Khanh Dao Duc, Shira Faigenbaum-Golovin, Karen Habermann, Emmanuel Hartman, Christoph von Tycowicz, Chi Zhang, Wenjun Zhao, Felix Zhou. Learning the Geometry of Data: A Mathematical Review of Shape Space Analysis.

Preprint available here: <https://arxiv.org/abs/2606.17022>

[15] *Mary Chriselda Antony Oliver, Emmanuel Hartman, Tom Needham. Conic Formulations of Transport Metrics for Unbalanced Measure Networks and Hypernetworks. *Submitted to SIAM Mathematical Analysis*

Preprint available here: <https://arxiv.org/abs/2508.10888>

[14] Emmanuel Hartman, Nicolas Charon. SVarM: Linear Support Varifold Machines for Classification and Regression on Geometric Data. *Submitted to Journal of Machine Learning Research*

Preprint available here: <https://arxiv.org/abs/2506.01189>

[13] Emmanuel Hartman, Martin Bauer, Nicolas Charon. Self Supervised Networks for Learning Latent Space Representations of Human Body Scans and Motions. *Submitted to Journal of Mathematical Imaging and Vision*

Preprint available here: <https://arxiv.org/abs/2411.03475>

Conference Publications

[12] Emmanuel Hartman, Jose Eric Jaramillo, Charles Puelz, Tam Doan, Silvana Molossi, Monisha Akula, Dana Reaves-O’Neal, Shagun Sachdeva, and Nicolas Charon. Morphology-Based Clustering for Risk Stratification of Patients with Anomalous Aortic Origin of a Right Coronary. *Journal of the American College of Cardiology*

[11] Minglang Yin, Zan Ahmad, Emmanuel Hartman, Yashil Sukurdeep, Shiyi Chen, Jiwoo Noh, Nicolas Charon, David D Spragg, and Natalia A Trayanova. AI-Based Shape Analysis Correlates Left Atrial Appendage Morphology with Stroke Risk in Patients with Atrial Fibrillation. *Heart Rhythm*

[10] Emmanuel Hartman, Emery Pierson. VariGrad: A Novel Feature Vector Architecture for Geometric Deep Learning on Unregistered Data. *In Eurographics Workshop on 3D Object Retrieval (2023)*

Preprint available here: <https://arxiv.org/abs/2307.03553>

[9] Emmanuel Hartman, Emery Pierson, Martin Bauer, Nicolas Charon, Mohamed Daoudi. BaRe-ESA: A Riemannian Framework for Unregistered Human Body Shapes. *In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*

Preprint available here: <https://arxiv.org/abs/2211.13185>

[8] Emmanuel Hartman, Yashil Sukurdeep, Nicolas Charon, Eric Klassen, Martin Bauer. Supervised deep learning of elastic SRV distances on the shape space of curves. *In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, 4425-4433. (2021)

Preprint available here: <https://arxiv.org/abs/2101.04929>

Journal Publications

[7] *Karen Habermann, Emmanuel Hartman. Brownian Motion on Spaces of Discrete Regular Curves. *Electronic Communications in Probability*

Preprint available here: <https://arxiv.org/abs/2510.24593>

[6] Luís F. Pereira, Alice Le Brigant, Adele Myers, Emmanuel Hartman, Amil Khan, Malik Tuerkoen, Trey Dold, Mengyang Gu, Pablo Suárez-Serrato, and Nina Miolane. Learning from landmarks, curves, surfaces, and shapes in Geomstats. *Transactions on Mathematical Software*

Preprint available here: <https://arxiv.org/abs/2406.10437v1>

- [5] Jonathan Cerqueira, Emmanuel Hartman, Eric Klassen, Martin Bauer. Sobolev Metrics on Spaces of Discrete Regular Curves. *Discrete and Continuous Dynamical Systems (DCDS)*
Preprint available here: <https://www.aims sciences.org/article/doi/10.3934/dcds.2025086>
- [4] Emmanuel Hartman, Emery Pierson, Martin Bauer, Mohamed Daoudi, Nicolas Charon. Basis restricted elastic shape analysis on the space of unregistered surfaces. *International Journal of Computer Vision*
Preprint available here: <https://arxiv.org/abs/2311.04382.pdf>
- [3] Emmanuel Hartman, Martin Bauer, Eric Klassen. Square Root Normal Fields for Lipschitz surfaces and the Wasserstein Fisher Rao metric. In *SIAM Mathematical Analysis*
Preprint available here: <https://arxiv.org/abs/2301.00284>
- [2] Emmanuel Hartman, Yashil Sukurdeep, Eric Klassen, Martin Bauer, Nicolas Charon. Elastic shape analysis of surfaces with second-order Sobolev metrics: a comprehensive numerical framework. *International Journal of Computer Vision*, 131(5), 1183-1209. (2022)
Paper available here: <https://link.springer.com/article/10.1007/s11263-022-01743-0>
- [1] *Martin Bauer, Emmanuel Hartman, Eric Klassen. The Square Root Normal Field Distance and Unbalanced Optimal Transport. *Applied Mathematics and Optimization*, 85(3), 35. (2022)
Preprint available here: <https://arxiv.org/abs/2105.06510>

Software Packages

Pytorch_VariGrad. Pytorch implementation of the VariGrad feature vector architecture proposed in *VariGrad: A Novel Feature Vector Architecture for Geometric Deep Learning on Unregistered Data*. Available on Github at https://github.com/emmanuel-hartman/Pytorch_VariGrad

H2_SurfaceMatch. A toolbox for the comparison, matching and interpolation of triangulated surfaces with respect to a second order Sobolev metric. Includes a comprehensive statistical pipeline that allows for the computation of the Karcher means, tangent space principal component analysis, and motion transfer in the space of parametrized surfaces and in shape space. Available on Github at https://github.com/emmanuel-hartman/H2_SurfaceMatch

**BCMOUT-pytorch. An implementation of the block coordinate method introduced in *The Square Root Normal Field Distance and Unbalanced Optimal Transport* that operates on a larger set of manifolds and computes the Wasserstein-Fisher-Rao and Hellinger-Kantorovich distances. Available on Github at <https://github.com/emmanuel-hartman/BCMOUT-pytorch>

SphericalRieszSearch. An implementation of projected gradient descent to find minimal configurations of measures with large numbers of supports with respect to the Riesz energy. Available on Github at <https://github.com/emmanuel-hartman/SphericalRieszSearch>

WassersteinFisherRaoDistance. An implementation of the framework introduced in *The Square Root Normal Field Distance and Unbalanced Optimal Transport*. Available on Github at <https://github.com/emmanuel-hartman/WassersteinFisherRaoDistance>

supervisedDL-SRVFdistances. An implementation of the framework introduced in *Supervised deep learning of elastic SRV distances on the shape space of curves*. Available on Github at <https://github.com/emmanuel-hartman/supervisedDL-SRVFdistances>

* Author order reflects equal contributions by all listed authors

** Joint work with Dylan Connell as a part of the FSU directed reading Program

AWARDS

Best Poster at CIRM Conference on Geometric Sciences in Action	2024
Tam Family Travel Award Recipient	2024
Dwight B. Goodner Endowed Fellow	2023
Tam Family Travel Award Recipient	2022
Bettye Anne Busbee Case Graduate Fellowship Recipient	2021
Florida State Legacy Fellow	2018-2023
UNC Asheville Mathematics Department Top Math Student Award Recipient	2018
UNC Asheville Mathematics Department Scholarship Recipient	2018
UNC Asheville Honors Program Citizenship Award Recipient	2018
UNC Asheville Laurel Scholarship Recipient	2014-2018

ADVISING AND MENTORING

Murad Hossen - UH Graduate Student	2024-Present
Samundra Regmi - UH Graduate Student	2024-Present
Jose Jaramillo - UH Undergraduate Research	2024-2026
Sophie Allen - FSU Directed Reading Program	2023-2024
Samuel Vecchio - FSU Directed Reading Program	2023-2024
Dylan Connell - FSU Directed Reading Program	2022-2023

PRESENTATIONS

- Summer 2026, Parameterization Invariant Representations of Geometric Data for Learning Applications, *Auburn University Topology and Geometry Seminar* Auburn, Alabama
- Summer 2026, Static Formulations of Vector Valued Optimal Transport, *Math in Aquitaine* Bergerac, France
- Summer 2026, SVarM: regression and classification in the space of varifold representations of geometric data, *Curves and Surfaces* Saint-Malo, France
- Summer 2025, Studying the Geometry of Discretizations of Infinite Dimensional Manifolds, *Math in Umbria: Geometry, Shapes and PDEs* Città di Castello, Umbria, Italy
- Spring 2025, Parameterization Invariant Representations for Efficient Shape Learning, *Infinite-dimensional Geometry: Theory and Applications*, ESI Vienna, Austria
- Fall 2024, Geometric Deep Learning for Unregistered Shape Data, *SIAM Mathematics of Data Science* Atlanta, GA
- Summer 2024, Conic Gromov Wasserstein Distance for Unbalanced Measure Networks, *Math in Maine: A Workshop centered around Geometry, Shapes and PDEs*, Andover, Maine
- Summer 2024, Elastic Shape Analysis of Surfaces with Second-Order Sobolev Metrics, *CIRM Conference on Geometric Sciences in Action*, CIRM, Marseilles, France

Summer 2024, Geometric Deep Learning for Unregistered Shape Data *CIRM Conference on Geometric Sciences in Action*, CIRM, Marseilles, France (Won Best Poster Award)

Spring 2024, Numerical Frameworks For Elastic Shape Analysis Using Second Order Sobolev Metrics *AMS Special Session on the Geometry of the Shape Space* Joint Mathematics Meeting, San Francisco, CA

Fall 2023, Formulations of the Wasserstein-Fischer-Rao Distance and Their Implications *UF/FSU Topology and Geometry Meeting*, Tallahassee, FL

Fall 2022, H2.SurfaceMatch: A numerical framework for elastic shape analysis. *Shape Workshop at Universite Lille* Universite Lille, Lille, France

Spring 2022, SRNF Elastic Shape Distance as unbalanced optimal transport *SIAM Conference on Imaging Science* (Virtual)

Spring 2022, SRNF Elastic Shape Distance as unbalanced optimal transport *Florida State Mathematics Machine Learning Seminar* Florida State University, Tallahassee, FL

Spring 2022, A Numerical Framework for Shape Analysis Using Split Second-Order Sobolev Metrics *UF/FSU Topology & Geometry Meeting* University of Florida, Gainesville, FL

Spring 2022, A Numerical Framework for Shape Analysis Using Split Second-Order Sobolev Metrics *Florida State Mathematics Machine Learning Seminar* Florida State University, Tallahassee, FL

Fall 2021, A Numerical Framework for Shape Analysis Using H^2 Metrics *Florida State Statistics Shape Group Meeting* Florida State University, Tallahassee, FL

Fall 2021, Optimal Reparameterizations in the SRVF and SRNF *Florida State Mathematics Machine Learning Seminar* Florida State University, Tallahassee, FL

Fall 2021, SRNF Elastic Shape Distance as the Wasserstein-Fisher-Rao distance *Geometry and Topology meet Data Analysis and Machine Learning* (Virtual)

Summer 2021, Supervised Deep Learning Framework for Computing Distances on the Shape Space of Curves *6th International Workshop on Differential Geometry in Computer Vision and Machine Learning* (Virtual)

Spring 2021, An optimal transport approach to SRNF shape matching *Florida State Statistics Shape Group Meeting* Florida State University, Tallahassee, FL

Fall 2020, An algorithm for fast computation of shape distance in the SRNF framework. *Florida State Statistics Shape Group Meeting* Florida State University, Tallahassee, FL

Fall 2020, Precise computation of the SRNF shape distance. *Florida State Mathematics Machine Graduate Student Seminar* Florida State University, Tallahassee, FL

Fall 2020, A supervised deep learning approach for the computation of elastic SRV distances. *Florida State Mathematics Machine Learning Seminar* Florida State University, Tallahassee, FL

Summer 2020, Advanced Topics Exam Presentation. *Florida State Statistics Shape Group Meeting* Florida State University, Tallahassee, FL

Spring 2020, Deep Learning for Shape Analysis. *Florida State Statistics Shape Group Meeting* Florida State University, Tallahassee, FL

Spring 2020, Introduction to StyleGAN. *Florida State Mathematics Machine Learning Seminar* Florida State University, Tallahassee, FL

Spring 2018, Using Topology to Study Groups. *Twelfth Annual Undergraduate Math Conference* University of Tennessee at Knoxville, Knoxville, TN

Fall 2017, Building an Active Honors Student Administration. *2017 SRHC Conference* Southern Regional Honors Council, Asheville, NC

Fall 2016, Building an Active Honors Student Administration. *North Carolina Honors Association Fall 2016 Conference*, University of North Carolina at Greensboro, Greensboro, NC

CONFERENCES ATTENDED

Math in Aquitaine, Bergerac, France, 2026

Curves and Surfaces, Saint-Malo, France, 2026

SIAM Texas-Louisiana Sectional Meeting, Austin, TX, 2025

Math in Umbria: Geometry, Shapes and PDEs, Città di Castello, Umbria, Italy, 2025

Infinite-dimensional Geometry: Theory and Applications, ESI, Vienna, Austria, 2025

SIAM Mathematics of Data Science, Atlanta, GA, 2024

Geometric Sciences in Action, CIRM, Marseilles, France, 2024

Math in Maine: A Workshop centered around Geometry, Shapes and PDEs, Andover, Maine, 2024

AMS Southeast Sectional Meeting, Tallahassee, FL, 2024

Joint Mathematics Meetings, San Francisco, CA, 2024

CIS 25th Anniversary Workshop, Johns Hopkins University, Baltimore, MD, 2023

UF/FSU Topology & Geometry Meeting, Florida State University, Tallahassee, FL, 2023

Shape Workshop at Universite Lille, Universite Lille, Lille, France, 2022

Geometry, Topology and Statistics in Data Sciences, Institut Henri Poincaré, Paris, France, 2022

Non-Linear and High Dimensional Inference, Institut Henri Poincaré, Paris, France, 2022

SIAM Conference on Imaging Science, Virtual, 2022

UF/FSU Topology & Geometry Meeting, University of Florida, Gainesville, FL 2022

Geometry and Topology meet Data Analysis and Machine Learning, Virtual, 2021

6th International Workshop on Differential Geometry in Computer Vision and Machine Learning, Virtual, 2021

Twelfth Annual Undergraduate Math Conference, University of Tennessee at Knoxville, Knoxville, TN, 2018

2017 SRHC Conference , Southern Regional Honors Council, Asheville, NC 2017

North Carolina Honors Association Fall 2016 Conference, University of North Carolina at Greensboro, Greensboro, NC, 2016

North Carolina Honors Association Fall 2015 Conference, University of North Carolina at Asheville, Asheville, NC, 2015

Consortium for CS in Colleges: Southeastern Conference , Roanoke College, Salem, VA, 2015

PROFESSIONAL EXPERIENCE

Florida State University, Tallahassee, FL

August 2018- May 2024

Linear Algebra Instructor

January 2023- May 2023

Calculus III Instructor

May 2022- August 2022

Calculus II Instructor

January 2022-May 2022

Calculus I Instructor

May 2021- August 2021

Research Assistant

August 2020 - Present

Discrete Mathematics Recitation Instructor

May 2020 - July 2020

Precalculus Algebra Instructor of Record

August 2019 - April 2020

Business Calculus Teaching Assistant

June 2019 - July 2019

Linear Algebra Grader

January 2019 - April 2019

Trigonometry Teaching Assistant

August 2018 - December 2018

College Algebra Teaching Assistant

August 2018 - December 2018

French Broad River Academy, Asheville, NC

September 2016- May 2018

After School Computer Science Course Instructor

January 2017 - Dec 2017

Volunteer Teaching Assistant

September 2016 - May 2017

After School Tutor

January 2017 - May 2018

UNC Asheville Student Employee, Asheville, NC

September 2016- May 2018

Math Lab Assistant

September 2016 - May 2018

Computer Science Peer Tutor

September 2017 - May 2018

Freshman Peer Mentor

September 2017 - Dec 2017

UNDERGRADUATE RESEARCH

Order Theory Research, UNC Asheville

Jan 2017 - May 2018

Our group explored how certain configurations within a partial order affect the order dimension.

Specifically, we sought to show that every poset whose order dimension is greater than two contains a certain sub-order.

Topology and Group Theory Research, UNC Asheville

Jan 2018 - May 2018

This research project investigates a method of generating a topological space from a group. Furthermore, we used the topology generated by this method to prove facts about groups.

Computer Science Research, UNC Asheville

August 2016 - May 2018

This large scale software development project was an iteratively developed Unity implementation of a popular board game, King of Tokyo, using Object Oriented abstractions and Model-View-Controller architecture. I developed an Artificially Intelligent actor to compete against human players of the game and stayed on the project as the software architect for the final semester of development.

EXTRACURRICULAR ACTIVITIES AND SKILLS

Member of the Mathematics Graduate Student Council	March 2021-Present
President and Founding Member of the UNCA Student Honors Advisory Council	April 2017-May 2018
Founder of the Sparks Lecture Series through the UNCA Honors Program	November 2015-August 2017
Extensive Experience in: Java, Python, Processing, Mathematica, LaTeX, Matlab and C#	
UNC Asheville Dean or Chancellor's List	Fall 2014-Spring 2018
Volunteer at the Southern Regional Honors Conference	Spring 2017
Participant in the ICPC Mid-Atlantic USA Contest-2nd Place at UNC Chapel Hill	November 2016
Attendee of the North Carolina Governor's School West in Mathematics	Summer 2013
Volunteer Teaching Assistant at the French Broad River Academy	January 2017 - May 2017
Participant in the Consortium for CS in Colleges:SE Conference Programming Contest	November 2015
Volunteer at the NC Honors Association Conference	Fall 2015